

LECTURE 1: PROBABILITY SPACES AND MEASURES

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to elre@princeton.edu, thanks in advance!

E. Rebrova

Definition 1.1. A **probability space** is a triplet $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is an arbitrary set called the *sample space*, $\mathcal{F}$ is a σ -algebra, and $\mathbb{P}$ is a *probability measure*.

The precise definitions of a σ -algebra and a probability measure below formalize the intuition that:

- Ω is the set of *all possible outcomes* of an experiment.
- $\mathcal{F}$ denotes all *events that can be observed*. An event consists of multiple possible outcomes.
- $\mathbb{P}$ assigns numerical values (*probabilities*, or *likelihoods*) to all observable events.

SIGMA-ALGEBRAS ($\mathcal{F}$)

Let Ω be a set, and denote the *collection of all subsets* of Ω by 2^Ω .

Definition 1.2 (Sigma-algebra). A collection of subsets $\mathcal{F} \subset 2^\Omega$ is a **σ -algebra** on Ω if:

- (1) (Empty set is included) $\emptyset \in \mathcal{F}$.
- (2) (Closed under complement) For any set from σ -algebra $A \in \mathcal{F}$, its complement is included as well $A^c := \Omega \setminus A \in \mathcal{F}$.
- (3) (Closed under countable union) For any countable collection of subsets $A_i \in \mathcal{F}$, its union $\cup_i A_i \in \mathcal{F}$.

If we require that (3) only holds for *finite* subsets, then we call $\mathcal{F}$ an **algebra** of subsets. We call the sets in the algebra **observable**, or **measurable** subsets of Ω . This is because probability likelihood/measure will be defined for all these sets.

An easy question: why is any σ -algebra an algebra? Think through why it is natural to require (1)-(3) if we want to describe “all observable/possible events”. How many elements are in the smallest possible σ -algebra on Ω ?

A convenient trick to define a σ -algebra is to start from a given collection of events “that can definitely happen”, and extend to all possible countable unions and complements:

Definition 1.3 (Generated sigma-algebra). A σ -algebra generated by $\mathcal{A} \subseteq 2^\Omega$ is the smallest σ -algebra containing $\mathcal{A}$. Formally,

$$\sigma(\mathcal{A}) = \bigcap \{ \mathcal{G} \subseteq 2^\Omega : \mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{A} \subseteq \mathcal{G} \}.$$

A good exercise is to confirm that the definition above is well-posed: specifically, check that any (arbitrary) intersection of σ -algebras is itself a σ -algebra.

Remark 1.4. Different sets of generators can result in the same generated σ -algebra. For example, consider $\Omega = \{1, 2, 3\}$, and the generators $\{1\}$ and $\{2, 3\}$. However (and this is convenient!), in order to show that two generated σ -algebras coincide, it is enough to check that each of them contains the set of generators of the other.

Definition 1.5 (Borel sigma-algebra). If the set Ω is endowed with a topology,¹ then the **Borel σ -algebra** is the σ -algebra generated by the collection of all open sets, and is denoted by $\mathcal{B}(\Omega)$.

Definition 1.6 (Open sets in $\mathbb{R}$). We say that a set $U \subseteq \mathbb{R}$ is **open** if for every $x \in U$, there exists an ε -ball centered at x that belongs to U . (That is, for every $x \in U$, there exists $\varepsilon > 0$ such that for all $y \in \mathbb{R}$ such that $|x - y| < \varepsilon$, we have $y \in U$.) We say that a set $C \subseteq \mathbb{R}$ is **closed** if its complement $C^c = \mathbb{R} \setminus C$ is open.

Some key properties of open sets are that (arbitrary) unions of open sets are open, and finite intersections of open sets are open. In $\mathbb{R}$, examples of open sets include open intervals of the form (a, b) — i.e., excluding the endpoints — and their unions; and examples of closed sets include closed intervals of the form $[a, b]$.

The standard topology in $\mathbb{R}$ defined above easily generalizes to $\mathbb{R}^n$: we say that a set $U \subseteq \mathbb{R}^n$ is open if for every $x \in U$, there exists $\varepsilon > 0$ such that for all $y \in \mathbb{R}^n$ such that $\|x - y\| < \varepsilon$, we have $y \in U$ (here, $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^n$).

Exercise 1.7. Verify that all of the following alternative definitions of the Borel σ -algebra $\mathcal{B}(\mathbb{R})$ are equivalent:

$$\begin{aligned} \sigma(\{(a, b) : a < b \in \mathbb{R}\}) &= \sigma(\{[a, b] : a < b \in \mathbb{R}\}) = \sigma(\{(-\infty, b] : b \in \mathbb{R}\}) \\ &= \sigma(\{(-\infty, b] : b \in \mathbb{Q}\}) = \sigma(\{O \subset \mathbb{R} \text{ open}\}). \end{aligned}$$

(PROBABILITY) MEASURES ($\mathbb{P}$ AND μ)

Definition 1.8. Let $\mathcal{F}$ be a σ -algebra on Ω . A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is called a **probability measure** if

- (1) $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
- (2) $\mathbb{P}(A) \geq 0$ for any $A \in \mathcal{F}$.
- (3) $\mathbb{P}$ is a countably additive function; namely, for any countable disjoint collection of subsets $A_i \in \mathcal{F}$ (i.e., $A_i \cap A_j = \emptyset$ if $i \neq j$), we have $\mathbb{P}(\sqcup_i A_i) = \sum_i \mathbb{P}(A_i)$.²

¹Recall that a topology for Ω is a collection of subsets $\mathcal{X} \subset 2^\Omega$ such that $\emptyset, \Omega \in \mathcal{X}$; any arbitrary (finite or infinite) union of members of $\mathcal{X}$ belongs to $\mathcal{X}$; and any intersection of any finite number of members of $\mathcal{X}$ belongs to $\mathcal{X}$. Members of $\mathcal{X}$ are called open sets. For this class, it should be enough to focus on the standard topology on $\mathbb{R}$ and $\mathbb{R}^n$.

²We use the notation $A \sqcup B$ to denote the union of two disjoint sets A and B .

Think about the following: how can we interpret these defining properties in terms of the likelihood of events?

Remark 1.9. Some additional related definitions and notations:

- If we do not require $\mathbb{P}(\Omega) = 1$, then $\mathbb{P}$ is called a **measure**, which is typically denoted by $\mu : \mathcal{F} \rightarrow [0, \infty]$. Moreover, we say that $(\Omega, \mathcal{F})$ is a **measurable space**, and $(\Omega, \mathcal{F}, \mu)$ is a **measure space**.
- If $\mu(\Omega) < \infty$, the measure is called **finite**.
- If there exists a countable family of subsets A_i such that $\mu(A_i) < \infty$ and $\cup_i A_i = \Omega$, then the measure is called **σ -finite** (think of the real line, which is a union of countably many length one segments).
- If we reformulate (3) to involve only finite collections of disjoint sets, we say that μ is a finitely additive non-negative set-function.
- If we do not require μ to be non-negative, it is called a signed measure.

Some properties of (probability) measures. Note that if $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$. Indeed, $B = A \sqcup (B \setminus A)$ and $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$ from non-negativity and finite additivity of measures. This is called *monotonicity* of (probability) measures.

Exercise 1.10. Prove the following properties of probability measures:

- (Sub-additivity/union bound) If $A \subseteq \cup_i A_i$ then $\mathbb{P}(A) \leq \sum_i \mathbb{P}(A_i)$
- (Continuity from below) If $A_1 \subseteq A_2 \subseteq \dots$ and $\cup_i A_i = A$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$ as $n \rightarrow \infty$.
- (Continuity from above) If $A_1 \supset A_2 \supset \dots$ and $\cap_i A_i = A$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$ as $n \rightarrow \infty$.

Show that for arbitrary (not necessarily probability) measures $\mu : \mathcal{F} \rightarrow [0, \infty]$, the last property can fail but the first two still hold.

EXAMPLES OF PROBABILITY SPACES

Why do we define the probabilities on the events in $\mathcal{F}$ rather than on individual outcomes? *Why don't we always consider $\mathcal{F} = 2^\Omega$?* Actually, it is possible that (a) we do not want to, or (b) we cannot.

Let's first discuss (a). The σ -algebra represents the amount of information available, so we might want to emphasize the lack of information. For example, in a sequence of experiments of unspecified length, at time $t = t_0$, we only have information about the results of the experiments that have occurred up to t_0 (e.g., think of a sequence of throws of a die).

Example 1.11. Consider n coin tosses. Then the sample space is $\Omega_n = \{H, T\}^n$, which contains all possible sequences of n faces from the experiment. Any sequence can be observed (i.e., $\mathcal{F} = 2^{\Omega_n}$) and is equiprobable (i.e., $\mathbb{P}(\{w\}) = 2^{-n}$ for any $w \in \{H, T\}^n$). For any event $A \in \mathcal{F}$, we have $\mathbb{P}(A) = \sum_{w \in A} 2^{-n}$. Note that $\mathbb{P}, \Omega$ and $\mathcal{F}$ all depend on n .

What if we want to model a sequence of n tosses in one space, and allow the set of observable events to grow with time as we see more? We could fix $\Omega = \Omega_n$, and define a monotone sequence of σ -algebras $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_n$ with $\mathcal{F}_n = \mathcal{F}$ as follows. For $l < k$,

let $\mathcal{F}_l$ be defined by the following property: for any $A \in \mathcal{F}_l$, if $\omega = (\omega_l \mid \tau_{n-l})$ is in A , where ω_l denotes the outcomes of the first l coin tosses in the sequence ω and τ_{n-l} denotes the remaining outcomes in ω , then any other sequence $\omega' = (\omega_l \mid \tau'_{n-l})$ must also be in A for any other sequence τ'_{n-l} of $n-l$ coin tosses (i.e., ω' has the same first l outcomes, but the tail is allowed to be different).

The intuition is that $\mathcal{F}_l$ models the case where we do not know the outcomes beyond the l th experiment, and hence our observable events in $\mathcal{F}_l$ are not able to “distinguish” between the outcomes in the tail. Check that $\mathcal{F}_l$ satisfies the definition of a σ -algebra.

What if we do not know how many tosses will be made? If we do not know n , we can define $\Omega = \Omega_\infty$ to be the set of infinite (countable) sequences, and define the monotone sequence $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ of σ -algebras $\mathcal{F}_l$, $l = 1, 2, \dots$, as before, and consider the measurable space $(\Omega, \mathcal{F})$ with $\mathcal{F} := \sigma(\bigcup_{l \geq 1} \mathcal{F}_l)$.³ Note that in this case we do not have to know how many tosses were made exactly.

How can we define a probability measure on this space? Given any observable event $A \in \mathcal{F}_n$ after n coin tosses, what should the probability of A be? (For a more concrete example that we can formulate precisely using the definitions we have seen, consider the probability space $(\Omega, \mathcal{F}_n, \mathbb{P}_n)$ equipped with the uniform probability measure $\mathbb{P}_n$. Observe that even though the sample space is infinite, the σ -algebra $\mathcal{F}_n$ in this case is finite (think this through!). What is $\mathbb{P}_n(A)$?)

Example 1.12. We could also consider a countable sample space Ω and infinite σ -algebra. For example, this is the case for a random variable with the Poisson distribution, where we have, for example, $\Omega = \mathbb{N} = \{1, 2, 3, \dots\}$, $\mathcal{F} = 2^\Omega$, and $\mathbb{P}(\{k\}) = \frac{\lambda^k}{k!} e^{-\lambda}$. Note that in this case, individual outcomes cannot be equiprobable, otherwise the probability of the whole state space Ω must be infinite by countable additivity of the probability measure $\mathbb{P}$ (recall Definition 1.8).

³This is the cylindrical σ -algebra for the set of infinite binary sequences generated by the cylinder sets, which are the sets in which the outcome of a finite number of indices (i.e., coin tosses) are specified. Note that any individual outcome, say $\omega = \{T, T, T, \dots\}$, is not in any $\mathcal{F}_l$ for any $l \geq 1$, but $\omega \in \mathcal{F}$.